Amrita School of Engineering
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21MEE311-Finite Element Methods

Assignment 3. Part 1

AI-Augmented Finite Element Analysis - Assignment Series

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Problem Description

This assignment covers the development of a fully interactive, web-based Finite Element Analysis (FEA) educational software ("FEAScript"). The application features 3 distinct structural solvers: 1D Spring Elements, 1D Axial Bar Elements, and complex 2D Truss structures. The application allows users to define nodes, apply boundary constraints (fixed supports), apply nodal forces, and connect elements with specific material cross-sections. The solver computes and visualizes nodal displacements, reaction forces, and internal element stresses (tension and compression).

Assignment Result → https://feascripts.vercel.app

Project Files → https://github.com/p1ll3chan/FEAscript

AI Interaction Report

AI Tool Used: Antigravity , ChatGPT

AI Interaction Report (Prompts, errors, corrections)

- Prompts: The AI was prompted to convert traditional FEA mathematical formulas (local stiffness matrices, global assembly) into modular TypeScript code, and build a visually premium React interface. Further prompts directed the AI to build a 3D visualization canvas for the resulting deformations.
- Errors: Initial issues included incorrect TypeScript compilation configurations (`verbatimModuleSyntax` errors) and matrix dimension mismatches when assembling the global stiffness matrix across varying degrees of freedom (1 DOF for bars vs 2 DOF for trusses).
- Corrections: We corrected the build errors by updating the Vite and ESLint configurations. We also iterated heavily on the GUI aesthetics, briefly redesigning the entire landing page to mimic a minimalist real estate site before deciding to revert to the more readable, original glassmorphism dark-mode aesthetic. \

Validation & Engineering Interpretation using theory

The solvers were built strictly using the Direct Stiffness Method:

- Local stiffness matrices are generated per element (e.g., k = AE/L for axial bars).
- For 2D Trusses, a transformation matrix is used to rotate the local matrices into global coordinates based on the angle between nodes.
- Degrees of Freedom (DOF) corresponding to fixed boundary conditions are eliminated or penalized so the matrix equation [K]{u} = {F} can be solved for displacements {u}.
- The accuracy of computations was verified using Vitest framework unit tests (`spring.test.ts`, `bar.test.ts`, `truss.test.ts`), which cross-checked the code's output against known, hand-calculated textbook analytical solutions to ensure mathematical validity.

- **How to use the codes developed. How to implement the code ?**

- Implementation: The code is a standard Node.js/Vite project. Developers can clone the repository, run `npm install` to download dependencies (React, Three.js, mathjs), and start the application using `npm run dev`. It can also be easily deployed to hosting platforms like Vercel.
- Usage: When a user selects a solver, they use the interactive tables to Input Node coordinates. They then check boxes to apply fixed boundary conditions, input force magnitudes, and finally connect the nodes using Elements with defined material properties (Area, Young's Modulus). Clicking "Solve System" triggers the calculations.

- **What is the innovative step from your side?**

- Browser-Based Execution: Unlike traditional FEA software (ANSYS, Abaqus) which are massive desktop installations, this solver runs incredibly fast entirely within the user's web browser using JavaScript Math libraries without needing a backend server.
- Interactive Visualization: The integration of React Three Fiber renders dynamic 3D/2D views of the structure. When solved, the canvas automatically overlays the deformed shape over the original ghosted shape, and color-codes members in green (tension) and red (compression) for instant, intuitive engineering feedback.
- Educational Focus: The inclusion of an interactive "FEA manual" directly inside the application makes the tool highly accessible for complete beginners to structural analysis.

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